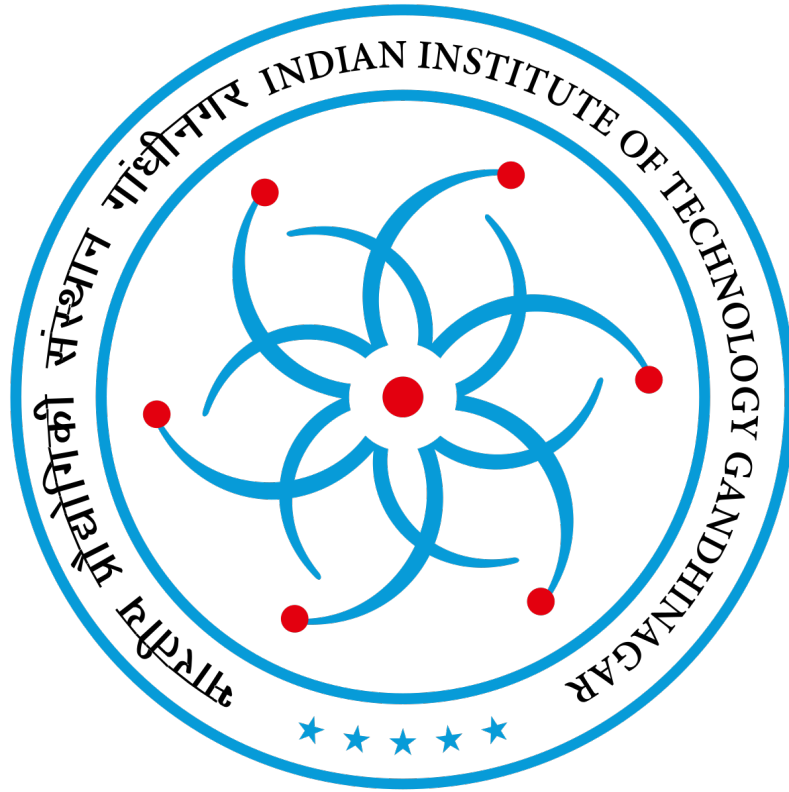


Indian Institute of Technology Gandhinagar



Fluid Flow through Converging-Diverging Rocket Nozzles

ME:605 Computational Fluid Dynamics

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Code Link: [Click here](#)

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1 Project Statement

1.1 Background and Assumptions

Nozzles are devices that are used to increase the velocity of the flow. They have numerous applications; one such application relevant to this project is rocket propulsion. In this case, the products of combustion are expanded through a convergent-divergent (CD) nozzle to extremely high velocities (typically supersonic speeds) to generate thrust.

The goal of this project is to computationally analyse fluid flow through a converging-diverging nozzle. To avoid excessive complexity, the following assumptions are invoked:

1. The flow is assumed to be quasi one-dimensional. In other words, flow properties are uniform across the cross-sectional area of the nozzle and therefore vary only along the axial direction.
2. The flow can be assumed to be inviscid, since the objective of the project is not to study the effect of walls on the fluid flow. Further, at high speeds, the convective terms in the governing equations are more important than the diffusive terms.
3. The fluid flow is assumed to be unsteady, laminar, and compressible.
4. The wall is assumed to be adiabatic. The radiative heat transfer and body forces are neglected.

2 Mesh Details

2.1 Mesh Type and Grid points

As we are simulating a quasi one-directional flow, we discretise the nozzle into n different points all lying in a single line. The number of nodal points in this computation is 51.

3 Nozzle Geometry

A converging-diverging nozzle is a classic one-dimensional compressible flow problem and is commonly used as a validation case in CFD code and software development. It illustrates key flow features such as choking, supersonic expansion, and shock formation within the nozzle. For this project, the converging-diverging verification nozzle case developed by the National Program for Application-Oriented Research in CFD (**NPARC**) is provided for study. This case was used to validate the WIND CFD code developed by NASA and AEDC, and details of those studies can be found here: [CVD Nozzle](#).

The nozzle geometry is described by the area (m^2) function as given below, and a schematic of the nozzle is shown in the figure below.

$$A = \begin{cases} 0.0444 - 0.019 \cos \left(\left(\frac{0.2x}{0.0254} - 1 \right) \pi \right), & \text{if } 0 \leq x < 0.127 \\ 0.0318 - 0.0063 \cos \left(\left(\frac{0.2x}{0.0254} - 1 \right) \pi \right), & \text{if } 0.127 \leq x \leq 0.254 \end{cases}$$

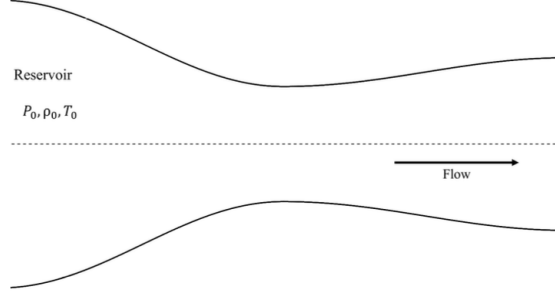


Figure 1: Converging-Diverging nozzle to be considered in the present study

4 Governing Equations

Since the flow is assumed to be unsteady, compressible, and inviscid, the time-dependent Euler equations must be solved. While the objective of the project is to compute the steady-state solution, the time-dependent Euler equations must be solved to ensure hyperbolicity of the governing equations across the entire Mach number regime of concern. The mass, momentum, and energy conservation equations for the quasi-one-dimensional flow are given below:

Continuity

$$\frac{\partial(\rho A)}{\partial t} + \frac{\partial(\rho AV)}{\partial x} = 0$$

where ρ is the density, A is the cross-sectional area, and V is the velocity.

Momentum

$$\frac{\partial(\rho AV)}{\partial t} + \frac{\partial(\rho AV^2)}{\partial x} = -A \frac{\partial P}{\partial x}$$

where P is the pressure.

Total energy equation

$$\frac{\partial}{\partial t} \left[\rho A \left(e + \frac{V^2}{2} \right) \right] + \frac{\partial}{\partial x} \left[\rho AV \left(e + \frac{V^2}{2} \right) \right] = - \frac{\partial(PAV)}{\partial x}$$

where e is the internal energy.

4.1 Conservative form of Governing Equations

The non-conservative form is highly susceptible to numerical instabilities and conservation issues and should be avoided in general. It is bound to fail in situations where there are shock waves in the solution domain, since the primitive variables change abruptly across the shock wave, while the conserved quantities remain unchanged. Hence, we solve the conservative form of the governing equations, only, and the governing equations in the conservative form can be simply represented as below:

$$\frac{\partial \mathbf{Q}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{J}$$

Here, Q , F and J are the following as shown below.

$$\mathbf{Q} = \begin{bmatrix} \rho A \\ \rho AV \\ \rho A \left(e + \frac{V^2}{2} \right) \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho AV \\ \rho AV^2 + pA \\ \rho AV \left(e + \frac{V^2}{2} \right) + pAV \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} 0 \\ p \frac{\partial A}{\partial x} \\ 0 \end{bmatrix}$$

To solve the above governing equations, all the fluxes (F) and source terms (J) also need to be represented in terms of conservative variables (Q). For that purpose, fluxes and source terms in terms of conservative variables are given below:

$$\mathbf{F} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} Q_2 \\ \frac{Q_2^2}{Q_1} + (\gamma - 1) \left(Q_3 - \frac{Q_2^2}{2Q_1} \right) \\ \frac{Q_2}{Q_1} \left[Q_3 + (\gamma - 1) \left(Q_3 - \frac{Q_2^2}{2Q_1} \right) \right] \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} 0 \\ J_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ (\gamma - 1) \frac{\partial(\ln A)}{\partial x} \left(Q_3 - \frac{Q_2^2}{2Q_1} \right) \\ 0 \end{bmatrix}$$

Here, γ is the specific heat ratio.

5 Boundary Conditions

The boundary conditions for the following problem have been shown below.

Table 1: Flow conditions in SI units.

Quantity	Value
Total Pressure at Inlet (Pa)	6894.76
Total Temperature at Inlet (K)	55.56
Exit Static Pressures (Pa)	6136.34 (Subsonic) 5171.07 (Shock) 1103.16 (Supersonic)

6 Methodology for Discretisation

6.1 MacCormack method

In computational fluid dynamics, the MacCormack method is a widely used discretisation scheme for the numerical solution of hyperbolic partial differential equations. This second-order finite difference method was introduced by Robert W. MacCormack in 1969. The MacCormack method is elegant and easy to understand and program.

MacCormack's technique is a variant of the Lax-Wendroff approach but is much simpler in its application. Like the Lax-Wendroff method, the MacCormack method is also an explicit finite-difference technique that is second-order accurate in both space and time.

The MacCormack's technique is a two-step process in which the first step is called the **predictor step** and the second step is called the **corrector step**.

6.1.1 Predictor Step

Let's consider our governing equation and discretise the terms for the predictor step. In the predictor step, we discretise the equations in Forward Time Forward Space (FTFS).

$$\frac{\partial \mathbf{Q}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{J}$$

At each grid point i :

$$Q_i^* = Q_i^n - \frac{\Delta t}{\Delta x} (F_{i+1}^n - F_i^n) + \Delta t J_i^n$$

- Q_i^* : Predicted value of conserved variable vector at the next intermediate step,
- F_i : Flux vector
- J_i : Source term vector
- n : current time step

The above discretised equation will be applicable for the terms Q_1, Q_2, Q_3 . In our governing equation, since the source terms J_1, J_3 are zero, they won't be present in the equation. The following discretised equations are shown below.

$$Q1_i^* = Q1_i^n - \frac{\Delta t}{\Delta x} (F1_{i+1}^n - F1_i^n)$$

$$Q2_i^* = Q2_i^n - \frac{\Delta t}{\Delta x} (F2_{i+1}^n - F2_i^n) + \Delta t J2_i^n$$

$$Q3_i^* = Q3_i^n - \frac{\Delta t}{\Delta x} (F3_{i+1}^n - F3_i^n)$$

After the conserved variable's predicted values have been calculated, it is important to calculate the updated values of the source terms vector and flux vector from the predicted values of the conserved variables.

$$\begin{aligned}
F_1^* &= Q_2^* \\
F_2^* &= \frac{(Q_2^*)^2}{Q_1^*} + (\gamma - 1) \left(Q_3^* - \frac{(Q_2^*)^2}{2Q_1^*} \right) \\
F_3^* &= \frac{Q_2^*}{Q_1^*} \left[Q_3^* + (\gamma - 1) \left(Q_3^* - \frac{(Q_2^*)^2}{2Q_1^*} \right) \right] \\
J_2^* &= (\gamma - 1) \frac{\partial(\ln A)}{\partial x} \left(Q_3^* - \frac{(Q_2^*)^2}{2Q_1^*} \right)
\end{aligned}$$

Now we have the predicted values of all the terms, and we can proceed to the corrector step.

6.1.2 Corrector step

This is the last step for the MacCormack's method. In the corrector step, we discretise the equations in the Forward Time Backwards Space (FTBS). We take the average of the predictor and corrected quantities to achieve second-order accuracy in both time and space.

$$\begin{aligned}
Q_{1,i}^{n+1} &= \frac{1}{2} \left[Q_{1,i}^n + Q_{1,i}^* - \frac{\Delta t}{\Delta x} (F_{1,i}^* - F_{1,i-1}^*) \right] \\
Q_{2,i}^{n+1} &= \frac{1}{2} \left[Q_{2,i}^n + Q_{2,i}^* - \frac{\Delta t}{\Delta x} (F_{2,i}^* - F_{2,i-1}^*) + \Delta t J_{2,i}^* \right] \\
Q_{3,i}^{n+1} &= \frac{1}{2} \left[Q_{3,i}^n + Q_{3,i}^* - \frac{\Delta t}{\Delta x} (F_{3,i}^* - F_{3,i-1}^*) \right]
\end{aligned}$$

Since $J_1 = J_3 = 0$ and only J_2 is nonzero:

$$J_{2,i}^* = (\gamma - 1) \left(Q_{3,i}^* - \frac{(Q_{2,i}^*)^2}{2Q_{1,i}^*} \right) \frac{d(\ln A)}{dx} \Big|_i$$

The last step is to calculate the updated values of the source terms vector and flux vector from the corrected values of the conserved variables at the new time step.

7 Isentropic subsonic flow

The signs of the eigenvalues of the problem help us determine the direction in which information propagates. The eigenvalues for our case are shown below. Here V is the speed of our wave and a is the speed of sound.

$$\lambda_1 = V - a, \quad \lambda_2 = V, \quad \lambda_3 = V + a$$

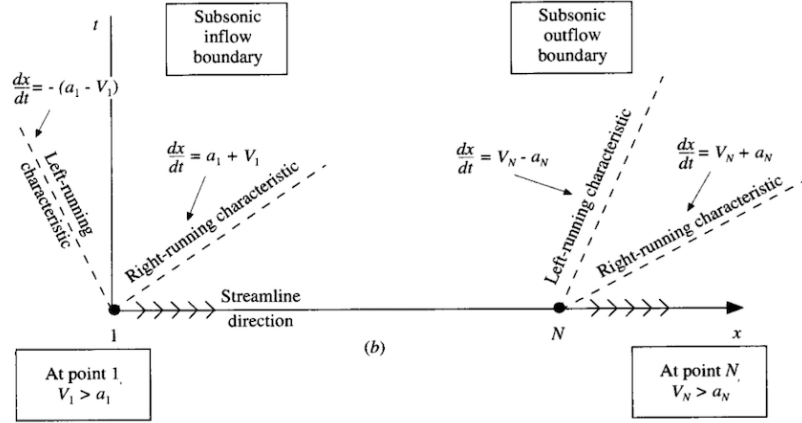


Figure 2: Characteristics for subsonic flow

Flow region	$M = V/a$	$\lambda_1 = V - a$	$\lambda_2 = V$	$\lambda_3 = V + a$
Subsonic	$M < 1$	< 0 (into domain)	> 0	> 0
Supersonic	$M > 1$	> 0	> 0	> 0

7.1 Boundary Conditions

At the subsonic inflow boundary, two properties are held fixed and one is allowed to float. At the outlet, the flow remains subsonic, meaning that two characteristics are flowing out of the domain; therefore, we can specify two thermodynamic properties at the inlet and one at the outlet. We choose to use linear extrapolation based on the flow-field values at the internal points.

At the inlet, the total Pressure and total temperature are known. So we extrapolate velocity using the following linear extrapolation.

$$V_1 = 2V_2 - V_3$$

The flow properties at the downstream, subsonic outflow boundary are obtained by linear extrapolation from the two adjacent internal points. If N denotes the grid point at the

outflow boundary, then we can calculate the density at the outlet since the back pressure is known.

$$\begin{aligned}(V)_N &= 2(V)_{N-1} - (V)_{N-2} \\ (T)_N &= 2(T)_{N-1} - (T)_{N-2}\end{aligned}$$

7.2 Initial Conditions

To initiate the computation, it is very important to specify the initial conditions along the nozzle. Since we are obtaining the steady-state solution, the initial conditions are merely a guess. The following initial guesses were taken for the project.

1. Linear function for Pressure at the interior nodes and the pressure at the first node to be equal to the Total inlet pressure, and the pressure at the last node to be equal to the outlet pressure.
2. Linear varying profile for velocity.
3. For a steady, one-dimensional, adiabatic (no heat transfer) flow through a nozzle, the Steady Flow Energy Equation simplifies to the conservation of **stagnation enthalpy** (h_0), or equivalently, **stagnation temperature** (T_0). So for the Initial guess for temperature, we use the following equation

$$\mathbf{T}_0 = \mathbf{T} + \frac{\mathbf{V}^2}{2c_p} = \text{constant}$$

4. We have estimated the initial pressure and temperature; then, using the ideal gas assumption, we can calculate the initial guess for density.

7.3 Time Step Calculation

As we are using an explicit scheme, which is vulnerable to numerical instability, it is crucial to calculate the correct time step to prevent the solution from blowing up.

The time step is governed by the CFL criterion, which requires that information cannot travel numerically faster than it does physically. The stability criterion states that the Courant number (C) must be less than or equal to 1. To calculate the time step, we use the following expression.

$$\Delta t = \min_i(\Delta t_i)$$

where the local time step is:

$$\Delta t_i = C \frac{\Delta x}{|V_i| + a_i}$$

- Δx : The constant grid spacing.

- V_i : The fluid velocity at grid point i .
- a_i : The speed of sound at grid point i .

7.4 Workflow of the Code

The working of the code has been explained below.

1. Define the necessary parameters and variables required for solving the problem.
2. Initialise the initial guess fields for temperature, velocity, pressure and density.
3. Initialise the Area function, Conservative variables, flux terms and source terms.
4. Create a loop in which we dynamically calculate the Δt using the time step condition.
5. In the same loop, we first start with the predictor step and using the discretised equation, we evaluate the predicted values of the conservative variables $Q1$, $Q2$, and $Q3$ for the interior nodes/
6. Using the predicted values, update the Sources terms and the Flux terms for the interior nodes.
7. We then proceed to the corrector step, in which we obtain the corrected values of the conservative variables $Q1$, $Q2$, and $Q3$.
8. Now we recover the primitive variables for the interior nodes so that we can extrapolate the values at the first and last node.
9. We apply the boundary condition required for the subsonic case, and we get the values for the primitive variables at the first and last node.
10. We update these values into the conservative variables $Q1$, $Q2$, $Q3$ which concludes the process for one time step.
11. Lastly, we calculate the residual, and the loop keeps going until the residual becomes less than the tolerance set by us.

7.5 Analytical Solution for isentropic flows

$$\frac{P}{P_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{\gamma}{\gamma - 1}}$$

$$\frac{\rho}{\rho_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{1}{\gamma - 1}}$$

$$\frac{T}{T_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1}$$

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2\right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

$$C = \sqrt{\gamma R T}$$

Here, A^* represents the throat area, and the subscript 0 refers to the reservoir state. C is the speed of sound.

7.6 Variation of Pressure Ratio along the nozzle

The variation of Pressure ratio along the nozzle, along with its comparison with the analytical solution, has been plotted below.

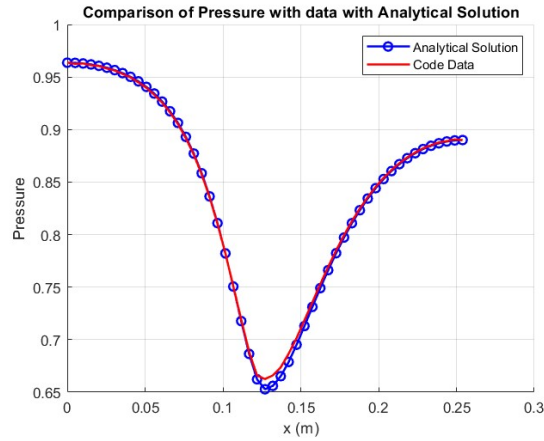


Figure 3: Variation of pressure ratio along the nozzle compared with analytical solution.

The explanation for the pressure ratio distribution for subsonic flow is as follows

1. In the case of Subsonic flow, the Mach number ($M < 1$), the change in density is small compared to the change in velocity. Therefore, to maintain a constant mass

flow rate, the velocity increases as the area decreases, and vice versa. According to Bernoulli's principle, as the velocity increases, the pressure decreases, and vice versa.

7.7 Variation of Density Ratio along the nozzle

The variation of density ratio along the nozzle, along with its comparison with the analytical solution, has been plotted below.

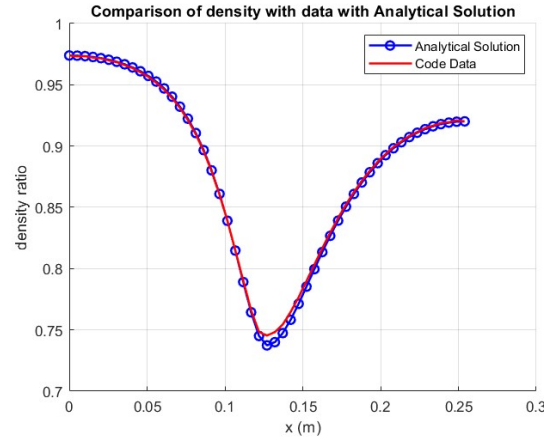


Figure 4: Variation of density ratio along the nozzle compared with analytical solution.

The explanation for the density ratio distribution for subsonic flow is as follows

1. In the case of Subsonic flow, the Mach number ($M < 1$), the change in density is small compared to the change in velocity. Therefore, to maintain a constant mass flow rate, the velocity increases as the area decreases, and vice versa. Hence, the increase in velocity is compensated by a decrease in the fluid density in the converging section and an increase in the density in the diverging section.

7.8 Variation of Temperature Ratio along the nozzle

The variation of the temperature ratio along the nozzle, along with its comparison to the analytical solution, is plotted below.

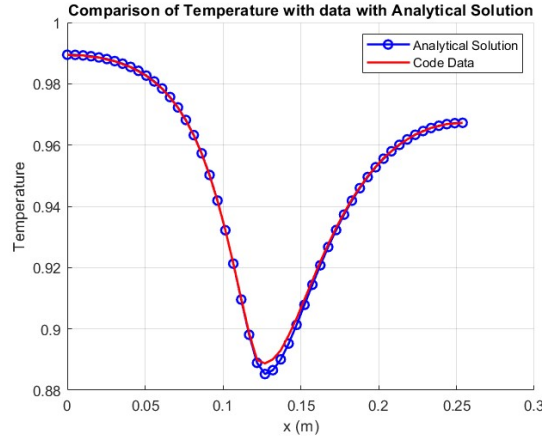


Figure 5: Variation of temperature ratio along the nozzle.

The explanation for the temperature ratio distribution for subsonic flow is as follows

1. For a steady, one-dimensional, adiabatic (no heat transfer) flow through a nozzle, the Steady Flow Energy Equation simplifies to the conservation of **stagnation enthalpy** (h_0), or equivalently. For an ideal gas, $h = c_p T$ and dividing by c_p gives the conservation of stagnation temperature:

$$h_0 = h + \frac{V^2}{2} = \text{constant}$$

,

$$\mathbf{T}_0 = \mathbf{T} + \frac{\mathbf{V}^2}{2c_p} = \text{constant}$$

2. We know that in subsonic flow, the velocity increases in the converging section and decreases in the diverging section. Since the total energy is constant, the increase in velocity must be compensated by a decrease in temperature, and vice versa. Hence, we get a decreasing temperature ratio profile in the converging section and an increasing temperature ratio profile in the diverging section.

7.9 Variation of Mach Number along the Nozzle

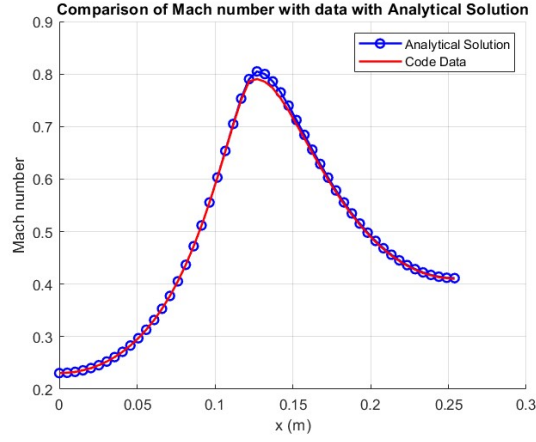


Figure 6: Variation of Mach Number along the nozzle.

The explanation for the Mach Number distribution for subsonic flow is as follows.

1. In the converging section, the flow remains Subsonic, where decreasing area increases the velocity, and hence the Mach number increases till the throat of the nozzle. In the diverging section, the flow remains subsonic, so increasing the area decreases the velocity, and hence the Mach Number decreases until the end of the nozzle.

7.10 Comparison of Pressure data with NASA's data along the nozzle

The variation of pressure along the nozzle, along with its comparison to NASA's data, is plotted below.

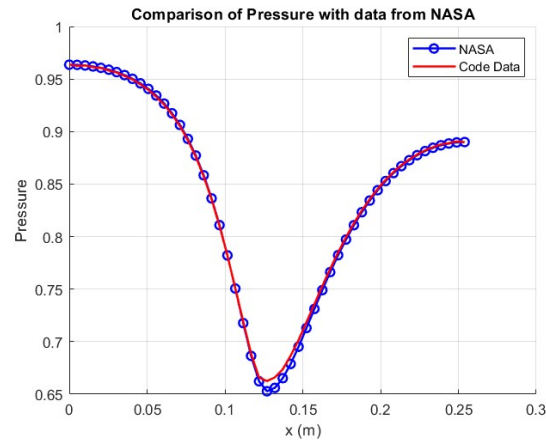


Figure 7: Variation of Pressure along the nozzle compared with NASA's data.

7.11 Comparison of Mach number with NASA's data along the nozzle

The variation of Mach number along the nozzle, along with its comparison to NASA's data, is plotted below.

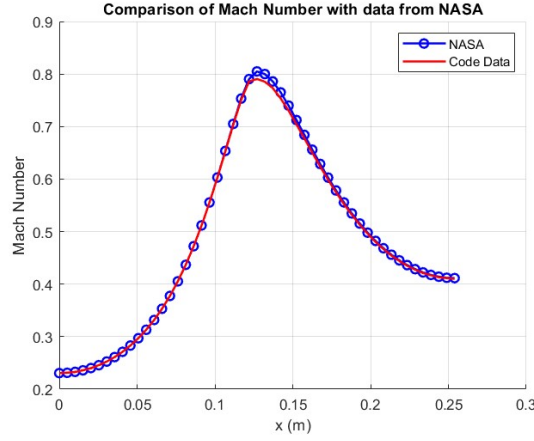


Figure 8: Variation of Mach number along the nozzle compared with NASA's data.

7.12 Main Reasons for Deviation at the Throat of the Nozzle

The deviation between the numerical data, analytical data and NASA's data near the nozzle throat is mainly due to the failure of the one-dimensional, inviscid flow assumption used in the code.

1. Viscous friction causes a reduction in entropy, which leads to a slight decrease in the stagnation pressure, which we have not considered in our code.
2. Near the throat of the nozzle, the flow is multidimensional, but we assume it to be one-dimensional, which leads to a deviation in the results.
3. We have also used a uniform grid; however, a more accurate solution can be obtained by increasing the number of grid points near the nozzle throat, as there is significant variation in properties in this region. This will enable us to capture all properties more accurately.
4. The viscous boundary layer effectively reduces the area available for the inviscid flow, which further increases the Mach number and reduces the pressure, density and temperature near the throat of the nozzle.
5. Not using Artificial viscosity results in sharp turns in the graph, which can be smoothed out by using artificial viscosity.
6. The current tolerance set is 10^{-6} but if we increase the tolerance to 10^{-10} the deviation will decrease.

8 Isentropic subsonic - supersonic flow

An isentropic subsonic-supersonic flow refers to a flow regime in which a compressible fluid transitions smoothly from subsonic speeds (Mach number $M < 1$ to supersonic speeds $M > 1$ without any increase in entropy.

The signs of the eigenvalues of the problem help us determine the direction in which information propagates and help us determine the number of properties that need to be defined at the inlet and the outlet. The eigenvalues for the governing differential equation are shown below. Here V is the speed of our wave and a is the speed of sound.

$$\lambda_1 = V - a, \quad \lambda_2 = V, \quad \lambda_3 = V + a$$

Flow region	$M = V/a$	$\lambda_1 = V - a$	$\lambda_2 = V$	$\lambda_3 = V + a$
Subsonic	$M < 1$	< 0 (into domain)	> 0	> 0
Supersonic	$M > 1$	> 0	> 0	> 0

8.1 Boundary Conditions

At the subsonic inflow boundary, two properties are held fixed, and one is allowed to float. In our case, we allow the velocity to float, and at the outlet, the flow has developed to supersonic, meaning that all three characteristics are flowing out of the domain; therefore, we cannot specify any thermodynamic property at the outlet. At the outlet, we allow velocity, temperature, and density to be floats and extrapolate their values using the interior nodes.

At the inlet, the total Pressure and total temperature are known, so we can evaluate the density. The only remaining term, which is velocity, is extrapolated using the following linear extrapolation.

$$V_1 = 2V_2 - V_3$$

The flow properties at the downstream, supersonic outflow boundary are obtained by linear extrapolation from the two adjacent internal points. If N denotes the grid point at the outflow boundary, then we can calculate the density at the outlet since the back pressure is known.

$$V_N = 2V_{N-1} - V_{N-2}$$

$$T_N = 2T_{N-1} - T_{N-2}$$

$$\rho_N = 2\rho_{N-1} - \rho_{N-2}$$

8.2 Initial Conditions

To initiate the computation, it is very important to specify the initial conditions along the nozzle. Since we are obtaining the steady-state solution, the initial conditions are merely a guess. The following initial guesses were taken for the project.

1. Linear function for Pressure at the interior nodes and the pressure at the first node to be equal to the Total inlet pressure, and the pressure at the last node to be equal to the outlet pressure.
2. Linear varying profile for velocity.
3. For a steady, one-dimensional, adiabatic (no heat transfer) flow through a nozzle, the Steady Flow Energy Equation simplifies to the conservation of **stagnation enthalpy** (h_0), or equivalently, **stagnation temperature** (T_0). So for the Initial guess for temperature, we use the following equation

$$T_0 = T + \frac{V^2}{2c_p} = \text{constant}$$

4. We have estimated the initial pressure and temperature; then, using the ideal gas assumption, we can calculate the initial guess for density.

8.3 Time Step Calculation

As we are using an explicit scheme, which is vulnerable to numerical instability, it is crucial to calculate the correct time step to prevent the solution from blowing up.

The time step is governed by the CFL criterion, which requires that information cannot travel numerically faster than it does physically. The stability criterion states that the **Courant number (C)** must be less than or equal to 1. To calculate the time step, we use the following expression.

$$\Delta t = \min_i(\Delta t_i)$$

where the local time step is:

$$\Delta t_i = C \frac{\Delta x}{|V_i| + a_i}$$

- Δx : The constant grid spacing.
- V_i : The fluid velocity at grid point i .
- a_i : The speed of sound at grid point i .

8.4 Workflow of the Code

The working of the code has been explained below.

1. Define the necessary parameters and variables required for solving the problem.
2. Initialise the initial guess fields for temperature, velocity, pressure and density.
3. Initialise the Area function, Conservative variables, flux terms and source terms.
4. Create a loop in which we dynamically calculate the Δt using the time step condition.
5. In the same loop, we first start with the predictor step and using the discretised equation, we evaluate the predicted values of the conservative variables $Q1$, $Q2$, and $Q3$ for the interior nodes/
6. Using the predicted values, update the Sources terms and the Flux terms for the interior nodes.
7. We then proceed to the corrector step, in which we obtain the corrected values of the conservative variables $Q1$, $Q2$, and $Q3$.
8. Now we recover the primitive variables for the interior nodes so that we can extrapolate the values at the first and last node.
9. We apply the boundary condition required for the supersonic case, and we get the values for the primitive variables at the first and last node.
10. We update these values into the conservative variables $Q1$, $Q2$, $Q3$ which concludes the process for one time step.
11. Lastly, we calculate the residual, and the loop keeps going until the residual becomes less than the tolerance set by us.

8.5 Variation of Pressure Ratio along the nozzle

The variation of Pressure ratio along the nozzle, along with its comparison with the analytical solution, has been plotted below.

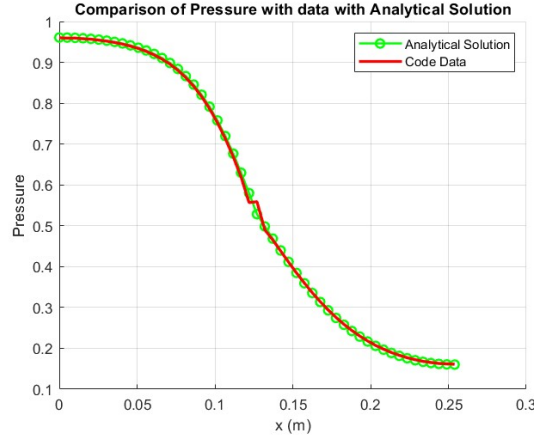


Figure 9: Variation of pressure ratio along the nozzle compared with analytical solution.

The explanation for the pressure distribution for subsonic-supersonic flow is as follows

1. The pressure drops steadily in the converging section of the nozzle where the flow remains subsonic ($M < 1$) owing to Bernoulli's principle, which says that as velocity increases, the pressure decreases as vice versa.
2. At the throat, the pressure reaches a value which is known as the critical pressure.
3. In the converging section, relationship between area (A) and velocity (V) is direct. As the flow area increases ($dA > 0$), the velocity continues to increase ($dV > 0$). This may seem counterintuitive, but it is necessary to satisfy the conservation of mass and momentum in highly compressible flow.

8.6 Variation of Density Ratio along the nozzle

The variation of density ratio along the nozzle, along with its comparison with the analytical solution, has been plotted below.

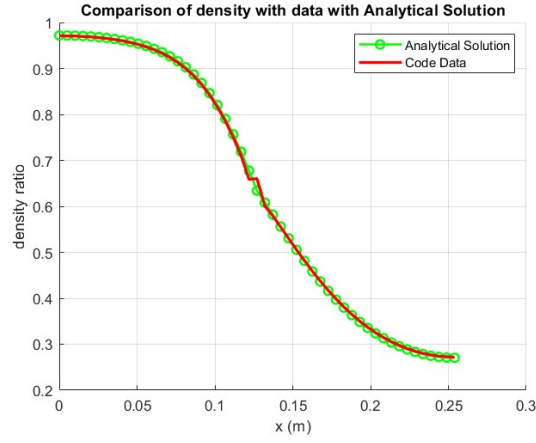


Figure 10: Variation of density ratio along the nozzle compared with analytical solution.

The explanation for the density distribution for subsonic-supersonic flow is as follows

1. As we assume the fluid to be an ideal gas, the density ratio variation follows a similar trend to the pressure ratio, since pressure is directly proportional to density; hence, the graph resembles the pressure ratio variation.

8.7 Variation of Temperature Ratio along the nozzle

The variation of the temperature ratio along the nozzle, along with its comparison to the analytical solution, is plotted below.

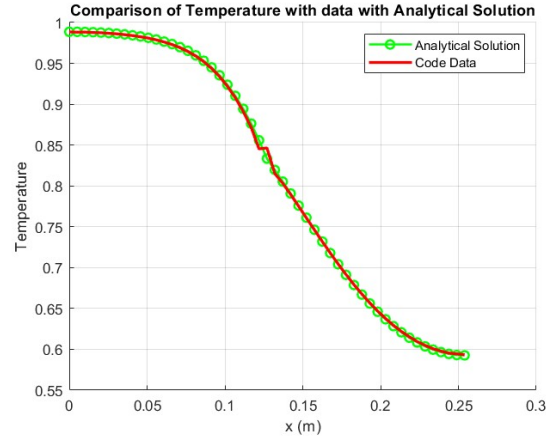


Figure 11: Variation of temperature ratio along the nozzle.

The explanation for the temperature ratio distribution for subsonic-supersonic flow is as follows

1. For a steady, one-dimensional, adiabatic (no heat transfer) flow through a nozzle, the Steady Flow Energy Equation simplifies to the conservation of **stagnation enthalpy** (h_0), or equivalently. For an ideal gas, $h = c_p T$ and dividing by c_p gives the conservation of stagnation temperature:

$$h_0 = h + \frac{V^2}{2} = \text{constant}$$

,

$$T_0 = T + \frac{V^2}{2c_p} = \text{constant}$$

2. We know that in supersonic flow, the velocity continues to increase even in the diverging section. Since the total energy is constant, the increase in velocity must be compensated by a decrease in temperature, and vice versa. Hence, we get a decreasing temperature ratio profile.

8.8 Variation of Mach Number along the Nozzle

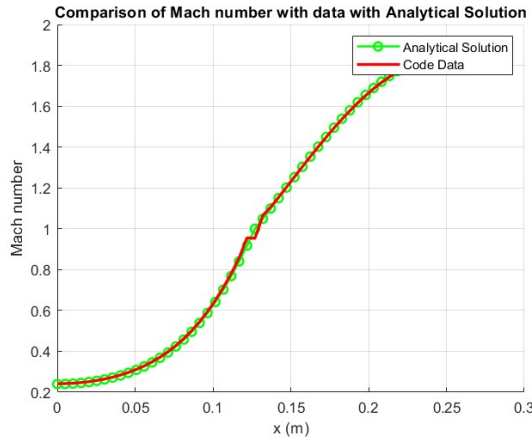


Figure 12: Variation of Mach Number along the nozzle.

The explanation for the Mach Number distribution for subsonic-supersonic flow is as follows.

1. In the converging section, the flow remains Subsonic ($M < 1$), though the flow is highly compressible; it behaves as an incompressible fluid, where decreasing area increases the velocity and hence the Mach number increases till the throat of the nozzle.
2. In the diverging section, the flow follows the inverse Area-Velocity relation, where expanding the area increases the velocity and thus the Mach number increases. A continuous drop in pressure and temperature drives this acceleration.

8.9 Comparison of Pressure data with NASA's data along the nozzle

The variation of pressure along the nozzle, along with its comparison to NASA's data, is plotted below.

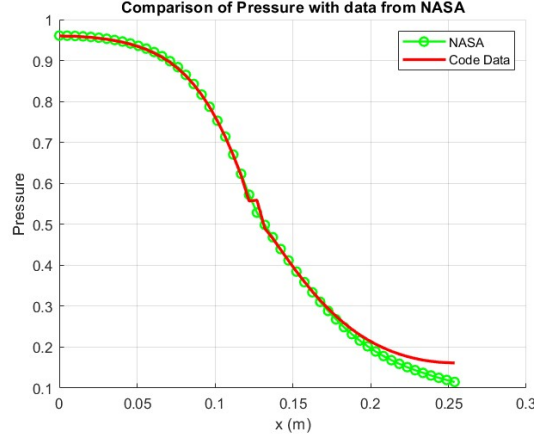


Figure 13: Variation of Pressure along the nozzle compared with NASA's data.

8.10 Comparison of Mach number with NASA's data along the nozzle

The variation of Mach number along the nozzle, along with its comparison to NASA's data, is plotted below.

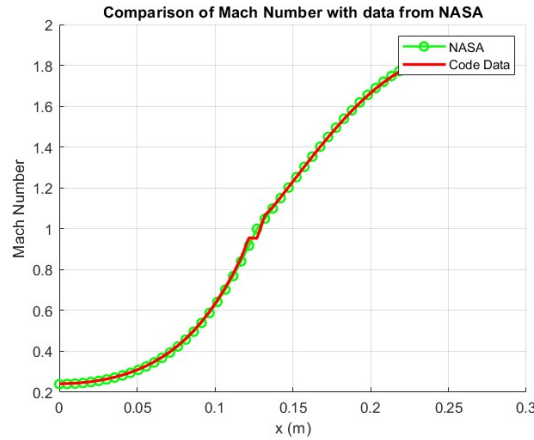


Figure 14: Variation of Mach number along the nozzle compared with NASA's data.

8.11 Main Reasons for Deviation at the Throat of the Nozzle

The deviation between the numerical data, analytical data and NASA's data near the nozzle throat is mainly due to the failure of the one-dimensional, inviscid flow assumption

used in the code.

1. Viscous friction causes a reduction in entropy, which leads to a slight decrease in the stagnation pressure, which we have not considered in our code.
2. Near the throat of the nozzle, the flow is multidimensional, but we assume it to be one-dimensional, which leads to a deviation in the results.
3. We have also used a uniform grid; however, a more accurate solution can be obtained by increasing the number of grid points near the nozzle throat, as there is significant variation in properties in this region. This will enable us to capture all properties more accurately.
4. The viscous boundary layer effectively reduces the area available for the inviscid flow, which further increases the Mach number and reduces the pressure, density and temperature near the throat of the nozzle.
5. Not using Artificial viscosity results in sharp turns in the graph, which can be smoothed out by using artificial viscosity.
6. The current tolerance set is 10^{-6} but if we increase the tolerance to 10^{-10} the deviation will decrease.

9 Results and Discussions

1. One of the most important steps before coding for any scheme is to check the signs of the eigenvalues, as they determine the direction in which the characteristics are flowing, meaning they help us determine the directions in which the information is propagating. Without a proper understanding of this, we cannot obtain results for any type of flow, whether subsonic or supersonic.
2. The next important thing is understanding the number of boundary conditions required at the inlet and outlet for each case, which can be understood only when we know the directions in which characteristics are flowing.

10 Concluding Remarks

This project helped me understand the numerical solution of hyperbolic PDEs and the numerical instability with the explicit scheme and provided insight into how the directions of characteristics affect the solution. I was able to understand why rocket nozzles are designed in a converging-diverging shape and why the velocity increases even in the diverging section in the supersonic case. Lastly, I was able to understand and solve the problem using MacCormack's Scheme.

11 Acknowledgments

I would like to express my gratitude to Professor Dilip Sundaram for teaching us the concepts of Computational Fluid Dynamics clearly and concisely, and for assigning this project to practically test our understanding of the concepts we were taught. Along with this, I would like to thank our Teaching Assistants, Mr Pardha Sai Mutyala and Mr Soham Pathak, and my friends for teaching me different algorithms during the tutorial sessions and helping me clarify any questions that arose while completing this project.

12 References

1. Computational Fluid Dynamics The Basics With Applications Anderson J D